

Thermal Hydraulics for Reactor Kinetics Lab

Selected architecture and sizing note

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1 Purpose

This note records the chosen thermal-hydraulic architecture and the first-pass sizing calculations that support it. It does three things:

1. explains why the current Modelica prototype overheats,
2. documents the operating envelope and loop topology that have been selected for the next implementation,
3. derives the reactor-scale flow, heat-removal, and pump targets that drive the corresponding Modelica design.

The note is intentionally close to the engineering-design level, not a safety case. Its output is a concrete Modelica subsystem breakdown ready for the next coding pass.

2 Why the current prototype overheats

The current Python neutronics model uses:

- nominal thermal power: $20 \text{ MW}_{\text{th}}$,
- annular active core: $R_{\text{inner}} = 0.8 \text{ m}$, $R_{\text{fuel}} = 3.456 \text{ m}$, $H = 6.912 \text{ m}$,
- heavy-water moderation (fixed assumption).

The current Modelica prototype is a small pedagogical loop. It:

- drives a single equivalent pipe on the order of lab-scale dimensions,
- uses a simple radiator for heat rejection,
- has no explicit saturation-margin constraint,
- imposes no reactor-appropriate pressure or flow boundary.

Because the primary loop is not sized to transport reactor-scale power at any physically meaningful flow rate, the numerical solver increases the hot-leg temperature until the enthalpy budget closes—in OMEdit this produces hot-leg readings reaching 400°C or beyond. This is a modelling artefact of the undersized loop, not a property of the reactor physics.

3 Reactor-scale inputs

The frozen geometry and power level from the repository are repeated here for reference:

Quantity	Value
Nominal thermal power P	20 MW
Inner moderator radius R_{inner}	0.80 m
Fuel outer radius R_{fuel}	3.456 m
Active height H	6.912 m
Primary coolant	D ₂ O

The active annular volume and average power density follow directly:

$$V_{\text{core}} = \pi(R_{\text{fuel}}^2 - R_{\text{inner}}^2)H \approx 245.5 \text{ m}^3, \quad q''' = \frac{P}{V_{\text{core}}} \approx 81.5 \text{ kW/m}^3.$$

This power density is low for a research reactor, which is consistent with the very large, heavily moderated annular geometry. The cooling challenge is therefore one of *flow capacity*, not heat-flux intensity.

4 Selected operating envelope

The following primary-loop temperatures and bulk temperature rise have been selected for the next Modelica implementation:

Quantity	Selected value
Primary inlet bulk temperature T_{in}	25°C
Nominal outlet bulk temperature $T_{\text{out,nom}}$	45°C
Upper-bound outlet temperature $T_{\text{out,max}}$	55°C
Nominal bulk temperature rise ΔT_{nom}	20 K
Upper-bound bulk temperature rise ΔT_{max}	30 K
Primary coolant flow direction	top-to-bottom
Primary pressure regime	low-pressure ($\lesssim 3$ bar gauge)

The inlet temperature of 25°C is set by the return temperature from the pool and the secondary heat exchanger. The nominal outlet of 45°C keeps the entire primary loop comfortably below saturation at any plausible low-gauge operating pressure and well within the single-phase D₂O regime. The 55°C upper bound is a conservative operating limit, not a safety margin; the actual saturation temperature at 3 bar gauge is approximately 133°C.

5 Reactor-scale flow and heat-removal estimates

5.1 Required primary mass flow

Using D₂O properties at the operating temperature midpoint ($\approx 35^\circ\text{C}$):

$$c_p \approx 4215 \text{ J/(kg K)}, \quad \rho \approx 1100 \text{ kg/m}^3.$$

The energy balance gives:

$$\dot{m} = \frac{P}{c_p \Delta T}.$$

Case	ΔT	$\dot{m}$ [kg/s]	$\dot{V}$ [m ³ /h]
Nominal design point	20 K	237	776
Upper-bound design point	30 K	158	518

The nominal design flow of approximately 237 kg/s (776 m³/h) is the primary target for pump and piping sizing. The upper-bound flow of 158 kg/s sets the lower flow limit at which the outlet temperature constraint is still met.

5.2 Pump duty

The pump shaft power scales with the loop pressure drop:

$$W_{\text{pump}} \approx \frac{\Delta p \cdot \dot{V}}{\eta}, \quad \eta \approx 0.7.$$

For the nominal design point:

Loop Δp	W_{pump} (nominal $\dot{m}$)	W_{pump} (upper-bound $\dot{m}$)
1 bar	31 kW	21 kW
2 bar	62 kW	41 kW
3 bar	92 kW	62 kW
5 bar	154 kW	103 kW

A total loop pressure drop in the range of 2–3 bar is a plausible first estimate for a large-scale research-reactor primary circuit, giving a pump duty on the order of 60–90 kW at the nominal flow.

5.3 Pool thermal time constant

The surrounding pool (Section 6) has a volume of 2535 m³ (2.535×10^6 litres). Its thermal time constant at full reactor power with no active cooling is:

$$\tau = \frac{m_{\text{pool}} c_{p,\text{water}}}{P} = \frac{2535 \times 10^3 \times 4200}{20 \times 10^6} \approx 530 \text{ s} \approx 9 \text{ min.}$$

This confirms that the pool cannot serve as the primary heat sink for sustained operation: at full power it would heat up by 20 K in roughly 9 minutes. The primary forced-circulation loop is therefore the essential heat-removal path, and the pool is a structural heat reservoir for short transients only.

6 Core, reflector, and pool abstraction

6.1 Actively cooled region

Forced primary D₂O flow is restricted to the fuel annulus and the inner moderator cylinder, i.e. the region from $r = 0$ to $r = R_{\text{fuel}} = 3.456$ m. The outer D₂O reflector ($R_{\text{fuel}} < r \leq R_{\text{refl}} = 4.056$ m) is not part of the forced-circulation path.

The detailed distribution of flow among individual fuel elements, coolant tubes, or sub-channels inside the active region is *not modelled*. Instead, the entire active region is represented by one **effective coolant channel** with:

- the total primary mass flow passing through it,
- the axially distributed core power profile as its heat input,
- an effective flow cross-section and hydraulic diameter chosen to give a representative pressure drop.

This single-channel abstraction is parameterised so that later refinement to multiple parallel channels (sharing inlet and outlet plena) can be done without restructuring the rest of the loop.

6.2 Flow direction

The primary coolant flows **top to bottom** through the effective core channel. The inlet is at the top of the active region; the outlet is at the bottom. This matches the natural downward-flow convention seen in several research-reactor designs and means the hottest bulk coolant leaves at the bottom before routing to the heat exchanger.

6.3 Stagnant reflector and pool

The outer D₂O reflector and the surrounding pool water together constitute a large stagnant inventory that provides:

- long-term thermal mass for decay-heat transient absorption,
- shielding of personnel and instrumentation,
- neutron moderation in the outer reflector zone.

The rectangular pool surrounding the reactor vessel has approximate dimensions of 15 m (height) $\times$ 13 m $\times$ 13 m (base), giving a total volume of approximately 2535 m³ (2.535×10^6 litres).

In the next Modelica model, this inventory is represented as a **fixed-temperature or slowly varying thermal boundary condition**. Active pool cooling, pool-loop flow, and pool temperature dynamics are not modelled in the first implementation pass.

7 Simplified primary loop

The primary D₂O loop follows a fixed topology inspired by forced-circulation research reactors such as FRM II, but simplified aggressively to the minimum useful set of components.

7.1 Topology

1. **Upper inlet plenum.** A vessel volume at the top of the core that distributes incoming primary coolant across the effective core channel entrance and provides hydraulic decoupling.
2. **Effective core channel.** A single axially discretised heated pipe (8 nodes) carrying the total primary flow top to bottom. Heat input follows the axial power profile from the neutronics model.
3. **Lower outlet plenum.** A vessel volume at the bottom of the core that collects the heated coolant before it exits to the primary piping.
4. **Primary pump.** A single centrifugal pump model with a representative head curve operating on the D₂O primary inventory. Nominal operating point: ≈ 237 kg/s at 2–3 bar head.
5. **Primary heat exchanger (primary side).** A counterflow shell-and-tube or plate-heat-exchanger model on the primary side that removes the full 20 MW to the secondary loop at nominal conditions.

7.2 Selected nominal operating parameters

Parameter	Value
Primary coolant	D ₂ O
Inlet temperature	25°C
Nominal outlet temperature	45°C
Upper outlet limit	55°C
Nominal mass flow	237 kg/s
Nominal volumetric flow	≈ 776 m ³ /h
Loop pressure regime	≲ 3 bar gauge
Core flow direction	top-to-bottom

8 Simplified secondary loop

The secondary side removes heat from the primary loop and rejects it to an ultimate heat sink. It is modelled at the minimum useful resolution:

1. **Secondary heat exchanger (secondary side).** The secondary-side complement of the primary HX above; coupled thermally via a fixed or parameterised overall heat-transfer coefficient UA .
2. **Secondary pump.** A single pump circulating light-water (H₂O) through the secondary side.
3. **Feed tank / source boundary.** A cold-water source at approximately 20°C representing the secondary supply.
4. **Sink tank / discharge boundary.** A downstream boundary that receives warmed secondary water and represents the ultimate heat sink without modelling it in detail.

There is no tertiary loop in this model. The secondary side is not intended to represent a real balance-of-plant; it exists to close the energy balance and allow transient study of primary-to-secondary coupling.

9 Modelica implementation targets

The next Modelica thermal-hydraulic model should consist of the following subsystems connected in a closed primary loop and an open secondary loop:

Subsystem	Modelica component class	Notes
Upper inlet plenum	<code>Fluid.Vessels.ClosedVolume</code>	top of core
Effective core channel	<code>Fluid.Pipes.DynamicPipe</code>	8 nodes, top-to-bottom
Lower outlet plenum	<code>Fluid.Vessels.ClosedVolume</code>	bottom of core
Primary pump	<code>Fluid.Machines.Pump</code>	head-curve parameter
Primary HX (primary side)	<code>Fluid.HeatExchangers</code>	counterflow
Secondary HX (secondary side)	coupled to primary HX above	
Secondary pump	<code>Fluid.Machines.Pump</code>	light-water side
Secondary source	<code>Fluid.Sources.FixedBoundary</code>	20°C supply
Secondary sink	<code>Fluid.Sources.FixedBoundary</code>	downstream boundary
Pool/reflector	<code>Thermal.HeatTransfer.Sources.FixedTemperature</code>	25°C

9.1 Key scaling parameters

The following parameters must be set to reactor scale, not to laboratory scale, in the next model:

- **Nominal power:** 20 MW (currently prototype uses ~ 10 kW).
- **Primary nominal flow:** 237 kg/s.
- **Pipe diameter and cross-section:** sized so that the channel velocity and pressure drop are physically plausible at the nominal flow (not the current ~ 10 mm toy pipe).
- **HX heat-transfer area:** sized to remove 20 MW with a primary-to-secondary temperature approach of ≈ 10 –15 K.

9.2 FMI variable exchange

The minimum set of variables to exchange at the Modelica–Python FMI interface:

Direction	Signal	Description
Python $\rightarrow$ Modelica	Q_total	total fission power [W]
Python $\rightarrow$ Modelica	q_axial [n]	normalised axial power fractions
Modelica $\rightarrow$ Python	T_in	core inlet bulk temperature [K]
Modelica $\rightarrow$ Python	T_out	core outlet bulk temperature [K]
Modelica $\rightarrow$ Python	m_dot	primary mass flow rate [kg/s]
Modelica $\rightarrow$ Python	dp_core	core pressure drop [Pa]

10 What is deliberately out of scope

The following items are excluded from the first implementation pass:

- Multiple parallel primary pumps or parallel heat-exchanger trains.
- Detailed in-core flow distribution or hot-channel analysis.
- Outer reflector cooling flow.
- Active pool cooling loop.
- Pool temperature dynamics (pool treated as a fixed boundary).
- Tertiary heat-rejection loop.
- Two-phase flow or void fraction modelling.
- Decay-heat tracking after shutdown.
- Safety system actuation or emergency core cooling.
- Full balance-of-plant or turbine cycle.

These items are noted here so that the implementation boundary is explicit and so that future extensions can be planned against this baseline rather than discovered incrementally.